Measurement in Physical Education and Exercise Science — Original Article

Section: Exercise Science

How much is the wind worth? Sprints and horizontal jumps from age 10 to senior level

Atle G. Guttormsen[1,*] (ORCID: 0000-0003-0188-8462)

Simen Guttormsen[1] (ORCID: 0009-0002-2533-8561)

[1]School of Economics and Business, Norwegian University of Life Sciences (NMBU), Ås, Norway

*Corresponding author: Atle G. Guttormsen, School of Economics and Business, Norwegian University of Life Sciences (NMBU), P.O. Box 5003, 1432 Ås, Norway. E-mail: atle.guttormsen@nmbu.no

Running head: Wind effects across age and ability

Word count: ∼5 500 (main text, excluding abstract, references, tables, and figure captions)

Abstract

Almost all wind-effect estimates come from elite seniors, yet the ±2.0 m/s legality rule applies equally to ten-year-olds. Using the Norwegian Athletics Federation's registry, we estimated wind effects within athlete-seasons in sprints and horizontal jumps at ages 10–34 (392 797 results). Per m/s of wind, results changed by 0.059 s on 60 m, 0.082 s on 100 m, 0.121 s on 200 m, 3.2 cm in the long jump and 4.2 cm in the triple jump. Absolute sprint effects declined with age (200 m, males: 0.144 s per m/s at 10–11 to 0.085 at 20–34) and ability, scaling approximately with race duration. Headwinds cost more than tailwinds gave, on the 100 m by about as much as drag physics predicts at athletes' own speeds. A legal +2.0 m/s tailwind was worth 0.11–0.25 s on the 100 and 200 m from age 12. Age-specific corrections can make lists, records and selection fairer.

Keywords: wind; sprint; long jump; youth athletics; performance measurement

1 Introduction

Wind is the one weather variable athletics regulates. Sprint and horizontal-jump results count for records only when tailwind does not exceed 2.0 m/s, gauge readings are part of the official result, and seasonal lists in many countries separate legal from wind-aided marks. How much the wind actually moves a result has been estimated in two ways: physical and quasi-physical models of the sprint (Linthorne, 1994; Mureika, 2001), and data-driven analyses of elite competition results (Moinat et al., 2018; Hollings et al., 2012). Both traditions agree on the order of magnitude for elite seniors — roughly 0.04–0.06 s per m/s in the 100 m, somewhat more for women than for men — and both concern elite seniors only. The one population study of non-elite sprinters we know of, on US high-school 100 m results, found that headwinds slowed girls and boys more than a drag model for elite men predicts and attributed the excess to psychology (Lybbert and Zheng, 2023); it did not examine how the effects scale with age or ability. Reviews of sprint testing likewise treat wind as a condition to standardize, with correction factors derived from adult data (Haugen and Buchheit, 2016). In measurement terms, a mark is meant to measure the athlete but also measures the conditions: wind is a known and recorded source of construct-irrelevant variance in sprint and jump marks, and its size at each age determines how far marks from different races can be compared.

Yet nothing guarantees that elite estimates transfer down the age and ability range, and the physics pulls in two directions. A slower runner spends longer on the course — a 12-year-old needs roughly a third more time over 100 m than a senior — so any given wind acts on them longer, inflating the absolute time effect. But aerodynamic drag scales with the square of airspeed, so slower runners spend a smaller share of their power against air, deflating it; children's higher surface-to-mass ratio pushes the other way again. The net scaling of the wind effect across age and ability is an empirical question that elite data cannot answer, and it matters practically: age-group records, seasonal youth lists, age-graded scoring tables and qualification

standards all treat marks within ±2.0 m/s as equivalent, in every age class, and talent evaluation leans on exactly these marks (Guttormsen, 2026).

The Norwegian registry is unusually well placed to answer it. Every sanctioned result in Norway is registered, and wind is measured and recorded down to the outdoor 60 m sprints that children contest from age 10 — to our knowledge the only population-scale source of wind-measured sprint data on children. We pre-specified five questions: the direction of the effect within athlete and season (H1, a design check); how the absolute effect scales with age (H2 — the exposure-time and aerodynamic-share mechanisms predict opposite directions); how the percentage effect scales (H3); whether headwinds cost more than tailwinds give (H4, as aerodynamics predicts); and what the ±2.0 m/s window is worth in each class, expressed against athletes' own performance variability (H5). The analysis plan was committed to version control before estimation; the study was not externally preregistered. After the main estimations, by a post-plan decision, class-specific short hurdles were added as a documented extension, reported in the supplementary material.

2 Methods

2.1 Data

We used a frozen extract (29 October 2026) of the Norwegian Athletics Federation's results database, which holds results electronically from 2011 (with thinner coverage in 2011, the first electronic season). The main analysis covers all results from 2011 onwards in nine wind-measured events: 60 m, 80 m, 100 m, 150 m and 200 m sprints, and long jump, long jump from a take-off zone, triple jump and triple jump from a zone (in the zone variants, used in the youngest classes, take-off is from a marked zone rather than a board, which reduces fouls). Only approved, outdoor, electronically timed results of athletes with known sex and birth year

were analyzed, at calendar-year ages 10–34, with event-specific plausibility bounds and wind readings within ±8 m/s. Before these filters we removed duplicate records created by re-imports and by meets stored twice, and excluded the board-event jumps at ages 11–13 in 2026, because that year's import filed take-off-zone jumps under the board events (details and counts in the Supplementary Methods). Of 670 803 extracted rows in the main events, 392 797 entered the analysis. The largest exclusions were indoor results — principally the indoor 60 m (86 449 rows), 200 m (35 849) and long jump (25 862) — and 16 246 outdoor rows without a wind value (overall, 4.8 % of approved outdoor results lack a reading). Missing wind is a completeness rather than a selection concern: it is concentrated in the smallest meets (9.0 % of rows in the smallest third of meets by size, against 1.4 % in the largest third) and its yearly share fluctuates between 2 and 7 %. Of the analyzed rows, 295 293 lie in athlete-season-event cells with at least two wind-measured results and identify the estimates. Times are analyzed in seconds and jumps in meters (reported in centimeters); units were verified against known magnitudes per event.

The same estimator was then applied, after the main analyses, to two external samples of wind-measured outdoor results in the 100 m, 200 m, 100 and 110 m hurdles, long jump and triple jump: all such results in the TFRRS profiles of NCAA Division I athletes, 2008–2026, read from a cleaned, hash-checked shared build of the profiles (546 854 results of 36 540 athletes after the registry's filters, with combined-event sub-results excluded), and the career results, as listed by Tilastopaja, of athletes who reached a World Championships, Olympic or other major championship final between 1983 and 2026 (66 589 results of 2 219 athletes). Neither source gives every athlete's age, so they are compared with the registry at their mean race times, the collegiate sample also by ability tertile.

2.2 Statistical model

The identifying variation is within athlete × season × event: the same athlete, in the same event and season, racing under different winds. For each event we estimated

$$y_{it} = \alpha_{i(\text{athlete}\times\text{year})} + \beta \cdot \text{wind}_{it} + \text{month dummies} + \text{round dummies} + \varepsilon_{it},$$

by within-demeaning; 74 110 athlete-season-event cells contribute at least two wind-measured results, and estimates are reported only where at least 300 results in 100 athlete-seasons contribute. Because the gauge reading is constant within a race, the race — and conservatively the meet — is the natural clustering level (Moulton, 1986); standard errors are therefore clustered on meet, counting the absorbed athlete-season effects in the small-sample correction because they are not nested within meets, with athlete clustering (the analysis plan's original choice, which gives intervals roughly half as wide in the large sprints and somewhat narrower ones elsewhere) retained as robustness. The month dummies are pooled across years and capture a common within-season profile; year-specific shocks are absorbed only in the meet fixed-effects specification. β is reported per m/s of wind, signed as a gain with tailwind, in seconds or centimeters, and in percent from a log-outcome specification; because headwinds cost more than tailwinds give, this linear slope averages the two sides, and statements about tailwinds use the separate tailwind slope. Age-band (10–11 through 20–34) and sex-specific models address H2/H3; ability tertiles address ability scaling, with ability defined as the athlete-season mean of *wind-adjusted* performance (each mark corrected to still air with the event-level slope), so that the ability measure itself is not contaminated by wind exposure, and the contributing athlete-seasons divided into equal thirds within event × sex × age band; separate slopes for head- and tailwind address H4; a quadratic term probes curvature. For H5, the advantage of a +2.0 m/s tailwind over still air is twice the slope over 0 to +2 m/s, estimated for each event, sex and

age band in a piecewise model with separate slopes for headwinds and for tailwinds above +2 (the response flattens above +2, so a single tailwind slope would understate the window, and the symmetric slope would overstate it). It is expressed in seconds and as a share of a robust within-athlete-season scale: 1.4826 times the median absolute deviation of within-cell residuals of wind-adjusted marks, over cells with at least three results and corrected for the estimated cell means. The robust scale is needed because the event-wide plausibility bounds admit a few gross recording errors (for example 200 m times filed as 100 m), which barely move the slopes but inflate a classical standard deviation. Pre-specified robustness analyses required at least three results per cell, excluded readings of exactly 0.0, split the period (2011–2019 vs. 2021–2026; 2020, whose season was curtailed by the pandemic, belongs to neither half), and replaced month dummies with meet fixed effects (two-way demeaning); a post-plan check excluded readings within ± 0.2 of +2.0. Other departures from the analysis plan are listed in the Supplementary Methods. Analyses used Python 3.13.7 with pandas 2.3.3, numpy 2.4.1 and statsmodels 0.14.6.

Two interpretive rules were fixed in advance. First, a single gauge reading does not measure the wind each athlete met; to the extent this error is classical it attenuates estimates toward zero, so the sprint effects are read as lower bounds. The measurement protocol also differs by event family: under World Athletics rules the gauge beside the straight averages 10 s from the gun in the 100 m (13 s in the short hurdles), 10 s from when the leader enters the straight in the 200 m — so it measures the wind over only part of a course that is half bend — and 5 s per attempt in the jumps; Norwegian practice applies the same conventions to the domestic distances (60 and 80 m as the 100 m; 150 m as the 200 m). Comparisons across event families bear this in mind. Second, wind is not randomly assigned: identification relies on within-season comparisons with month, round and (in robustness) meet controls, and no causal language beyond this design sense is used. For the jumps, one further caveat applies: the registry stores only the wind of

the measured (best) attempt, which describes the conditions of a series imperfectly, and the direction of the resulting bias is not established; the lower-bound reading is therefore applied to the sprints only.

2.3 Sample size

No sample size was calculated a priori, because the study used every result in the registry that met the inclusion criteria: the sample is the eligible population, and its size was fixed by the data rather than chosen (Lakens, 2022). An a posteriori sensitivity analysis (Supplementary Methods) puts the minimum detectable slope at 0.003 and 0.004 s per m/s in the 60 and 100 m, below the smallest effect of interest of 0.005 s per m/s (one recording unit of 0.01 s over the legal window), and higher in the smaller events and the thinner age bands; estimates are reported only where at least 300 results in 100 athlete-seasons contribute.

2.4 Ethics and transparency

The data are competition results registered by the Norwegian Athletics Federation: results from age 13 are publicly listed, while records for the 10–12 age classes are registered but not publicly released by the federation. No athlete is identified. Performance data below age 13 are used solely to estimate measurement conditions — not to rank, list or evaluate children — in keeping with the purpose of the Norwegian children's sport regulations; no result of an athlete under 13 appears in ranked, listed or identifiable form in the paper or its outputs. The study is not medical or health research as defined by Norway's Health Research Act (Helseforskningsloven §§ 2 and 4), so approval by a Regional Committee for Medical and Health Research Ethics was not required, and Sikt (the Norwegian Agency for Shared Services in Education and Research) confirmed that the use of the data raises no data-protection concerns. No participants were contacted or recruited, and informed consent was not required. The study was conducted

in accordance with the Declaration of Helsinki. The analysis plan was committed to version control before estimation; athlete-level files are regenerable from the registry. Reporting follows STROBE where applicable (von Elm et al., 2007).

3 Results

3.1 Sample and wind exposure

Table 1 describes the nine main events. Wind readings centered near calm with wide spread (median +0.3 m/s; 5th–95th percentile −2.1 to +2.7). The distribution shows digit preference — 8.7 % of readings are exactly 0.0, more than any continuous wind distribution would place there and consistent with some unoperated gauges being recorded as calm, and 16 % fall on whole meters per second — and thins just above the legality line: 3.4 % of readings lie in (1.8, 2.0] against 2.2 % in (2.0, 2.2], with 1.6 % exactly +2.0. Both patterns motivate robustness checks: excluding exact zeros, and excluding readings within ±0.2 of +2.0, leave every estimate essentially unchanged (Table S1). The bulk of the material is young: the 60 m, contested outdoors by the youngest classes, is the largest single event (103 286 analyzed results, of which 72 558 lie in athlete-seasons with at least two wind-measured races and thus contribute to estimation).

3.2 What a meter per second is worth (H1)

Table 2 reports the main within-athlete estimates. Per m/s of wind, results changed by 0.059 s (95 % CI 0.057–0.061) on 60 m, 0.082 s (0.080–0.085) on 100 m, 0.121 s (0.113–0.130) on 200 m, 3.2 cm (2.9–3.5) in the long jump and 4.2 cm (3.5–4.8) in the triple jump; in percentage terms, 0.39–0.68 % per m/s. All directions matched H1. The small 80 m and 150 m events gave consistent estimates with wide intervals.

The published elite figures of roughly 0.04–0.06 s per m/s on the 100 m (Linthorne, 1994; Moinat et al., 2018) mainly describe tailwinds. On the tailwind side alone, the population slopes were 0.047 s (60 m), 0.065 s (100 m) and 0.078 s (200 m) per m/s (Table S2), so the 100 m figure is about 1.1–1.6 times the elite range. Restricted to seniors (20–34), the 100 m tailwind slope was 0.064 s for men and 0.057 s for women, at or just above the elite range — and our senior men, averaging about 11.4 s on the 100 m, are themselves slower than the elite samples. The elite literature describes the fast end of a gradient. Applying the estimator unchanged to the two external samples places that end on the registry's scale (Table S3): on the 100 m, 0.054 s per m/s (95 % CI 0.053–0.056) for collegiate men and 0.058 (0.056–0.060) for collegiate women, 0.048 (0.045–0.051) and 0.059 (0.055–0.063) for championship finalists — 58–72 % of the registry's 0.082, and for three of the four groups within 0.002 s of the level formula of Moinat et al. (2018) at their mean times.

3.3 The effect falls with age — and with speed (H2, H3)

Figure 1 and Table 3 show the core result. In absolute seconds, the wind effect declined with age in every sprint and both sexes — monotonically in every series except the sparse girls' 200 m at 10–11: on 200 m for boys and men from 0.144 s per m/s at 10–11 to 0.085 at 20–34; on 60 m from 0.070 at 10–11 to 0.041 at 16–17; on 100 m from 0.101 at 12–13 to 0.071 at 20–34. Exposure time accounts for most of the gradient: mean race durations shrink in almost the same proportion (men's 100 m, 14.97 s at 12–13 vs. 11.42 s at 20–34, a ratio of 1.31, against a wind-effect ratio of 1.42), so the percentage effect (Figure 2) is far flatter — but not constant, declining for instance from 0.70 to 0.62 % per m/s on the men's 100 m and from 0.70 to 0.52 % on the 60 m. Dividing each event's effect by its mean race duration makes the scaling explicit (Table S4): the straight sprints line up at 0.0059–0.0063 s per m/s per second of race time, while the bend events sit a third lower (150 m 0.0040, 200 m 0.0043) — as expected when the

gauge, read beside the home straight, measures the wind over only part of a course that is half bend, and when the wind meets the runner at varying angles through the bend. In the jumps the absolute effect for girls and women ran the other way, rising with age in centimeters (long jump: 2.9 cm per m/s at 14–15 to 4.1 at 20–34) as jump distances and flight times grow, but not for boys and men (3.4 at 14–15, 3.1 at 20–34), and the percentage effect shows no consistent age direction (Table S5); below 14 nearly all horizontal jumps are made from a take-off zone, so the board-jump series start at 14–15. Within age bands, ability scaling follows the sprint pattern (Figure S1): with ability measured as the wind-adjusted season mean, the lowest tertile gained most per m/s in every sprint (100 m: 0.088 s against 0.079 in the highest tertile; 60 m: 0.066 against 0.056; 200 m: 0.133 against 0.117), while the jump tertiles did not separate. Slower athletes — whether young or simply weaker — are exposed longer, and no single correction factor, absolute or proportional, is exactly age-invariant.

Within the collegiate sample the same proportionality holds (its ability tertiles sit at 0.0047–0.0051 per second of race time on the 100 m), but across samples it does not: per second of race time the 100 m rate is 0.0062–0.0064 in the registry against 0.0047–0.0052 in the external samples, 15–27 % lower for the same sex, so the law of Table S4 predicts 0.061–0.077 s per m/s at their mean times against the observed 0.048–0.059 (Table S3). The gap persists at equal race times: the registry's seniors gained 0.071 s per m/s (0.066–0.075) against 0.057 (0.053–0.061) for the slowest collegiate tertile of men, at 11.4 and 11.3 s, and 0.072 (0.067–0.078) against 0.060 (0.057–0.064) for women, at 12.6 and 12.8 s. The per-second rate thus carries a sample-specific component — gauge practice, weather or population — that comparisons with elite estimates must allow for. A post-plan extension to the class-specific short hurdles (Supplementary Methods; Table S6 and Figure S2) found wind effects like those of the flat 60 m over the 60 m hurdles and mostly larger ones over the 100 and 110 m hurdles, which the hurdles' longer race times largely, though not entirely, account for.

3.4 Headwinds cost more than tailwinds give (H4)

In every large sprint event the headwind slope exceeded the tailwind slope (Figure 3; Table S2): on 100 m a -1 m/s headwind cost 0.107 s against a +1 m/s tailwind's 0.065 s ($p < 0.001$ for equal slopes); on 200 m, 0.184 against 0.078; on 60 m, 0.072 against 0.047. The positive quadratic terms tell the same story (200 m: +0.0141 s per $(\text{m/s})^2$, SE 0.0013; Table S7). Drag proportional to the square of airspeed predicts this direction, and on the 100 m also roughly this size. The familiar ratio $(2v + 1)/(2v - 1) \approx 1.14$–$1.18$ describes a wind of exactly ± 1 m/s, not slopes averaged over the observed range of winds; imposing drag-only physics (time change proportional to the change in squared airspeed at the athlete's mean speed, scaled so that its linear slope equals the observed one; only the head/tail ratio and the curvature relative to the slope are therefore predictions, and the mean speed, which includes the start, overstates the predicted asymmetry most on the 60 m) on the observed winds and fitting the same piecewise model predicts head/tail ratios of 1.84 (60 m), 1.72 (100 m) and 1.85 (200 m), against observed 1.52, 1.66 and 2.36, and quadratic terms of 0.0048, 0.0063 and 0.0099 against observed 0.0037, 0.0057 and 0.0141. The 100 m thus matches drag closely, the 60 m is somewhat less asymmetric, and only the 200 m — where the gauge reads only the home straight — is more asymmetric than drag alone predicts. On the 100 m the asymmetry was larger for girls and women than for boys and men (head/tail ratio 1.77, 95 % CI 1.58–1.96, against 1.54, 1.36–1.71), as drag predicts for slower runners (1.73 and 1.67 at their own speeds); in 9 of the 10 sex $\times$ age bands the drag-only ratio lay inside the observed interval, and the exception lay below it (Table S8). Within meets, where weather is held fixed, the ratios were 1.59 and 1.46 against drag-only 1.70 and 1.63. The collegiate 100 m ratios likewise matched drag at the athletes' own speeds (1.62 and 1.61 against 1.67 and 1.76, men and women), whereas the championship finalists exceeded it (1.93 and 1.86 against 1.45 and 1.49; Table S3); and in all three samples the tailwind slope above +2.0 m/s was 12–33 % lower than over 0 to +2 m/s,

against the 26–36 % that drag-only physics predicts for the observed winds, so the legality line leaves no mark on the response beyond physics. In the jumps the two slopes did not differ significantly in the long jump (2.93 against 3.71 cm, p = 0.18), and in the zone variants the tail-wind slope was the steeper one; since the registry stores only the wind of the measured attempt, the jump asymmetries are not interpreted.

3.5 What the legal window is worth in each class (H5)

Table 4 translates the estimates into the quantity the rules regulate: the advantage of a +2.0 m/s tailwind over still air, valued from the slope over 0 to +2 m/s (Section 2.2) and expressed in seconds and as a share of a robust measure of the athlete's own within-season variability. From age 12 the window was worth 0.11–0.25 s on the 100 and 200 m. For boys and men it showed no clear age trend in seconds: 0.14–0.16 s on the 100 m and 0.13–0.19 s on the 200 m in the bands from 12–13 to 20–34, with overlapping intervals. On the boys' 100 m the age decline of the linear slope (Section 3.3) lies mainly in the headwind slope (Table S9). For girls and women the window was larger before senior age, also with overlapping intervals: 0.15 s on the 100 m at 12–13 against 0.11 s at 20–34, and 0.25 against 0.14 s on the 200 m. As a share of race time it was smaller for boys aged 12–13 than for senior men on the 100 m (1.10 against 1.27 % of the mean time) and only slightly larger for girls than for senior women (0.95 against 0.87 %). Relative to the athlete's own variability the window is worth 0.29–1.26 of the robust scale across events, sexes and ages, and most among senior sprinters, whose marks scatter least (men's 100 m: 0.67 at 12–13 against 1.26 at 20–34). Two external yardsticks complete the picture. On the federation's age-graded scoring tables, +2.0 m/s on the 100 m converts to 22–29 points at age 13 against 17–25 points on the oldest (age-19) table, a youth margin of about the same proportion as in seconds. And in the 2025 national 100 m season lists (legal marks only, wind $\leq$ +2.0 m/s; senior lists open to all ages), correcting tailwind-aided marks to still air

with the class-specific slopes brings one new athlete into the top 10 of the girls' class-14 list and of the senior men's list and two into the senior women's list (none in the boys' class-14 list); in the senior lists 66 % (men) and 56 % (women) of top-50 marks carried tailwinds above +1 m/s: wind moves rankings at youth and senior ages alike.

3.6 Robustness

Estimates were essentially unchanged when cells required three or more results, when exact-zero readings were excluded, when readings within ± 0.2 of +2.0 were excluded, across period halves (except the triple jumps, whose 2021–2026 estimates were 14–24 % higher), and under athlete instead of meet clustering (Table S1; point estimates are identical under either clustering). Replacing month dummies with meet fixed effects — which compares results only within the same meet and absorbs any meet-level weather or gauge siting — attenuated the 60, 100 and 200 m estimates by 13–41 % (100 m: 0.082 to 0.066 s per m/s), left the small 80 and 150 m events essentially unchanged, and raised the jump estimates by 26–47 %. Only 25–34 % of the sprint wind variance lies within meets (39–51 % in the jumps, where attempts spread across the day), so the meet-FE estimates rest on far less variation. The sprint attenuation is compatible with gauge error that weighs more once between-meet variation is removed, but also with some between-meet confounding in the month-dummy estimates; these data cannot separate the two, and the rise in the jumps shows that meet-level factors can push either way. The age gradient survived meet fixed effects: in every sprint series the estimate fell from the youngest to the oldest well-measured band (men's 100 m: 0.107 s per m/s at 12–13 to 0.060 at 20–34; Table S10).

4 Discussion

4.1 Principal findings

In the first population-scale estimates of wind effects across age and ability, a meter per second of wind (linear fit over head- and tailwinds) was worth 42–70 % more, in absolute terms, to a boy sprinter than to a senior man — 0.10 versus 0.07 s on 100 m at 12–13, 0.14 versus 0.08 s on 200 m at 10–11 — and 34–47 % more among girls aged 12–13, with the published elite values (Linthorne, 1994; Mureika, 2001; Moinat et al., 2018; Hollings et al., 2012), and the 0.048–0.059 s per m/s that the same estimator returns in collegiate and world-class 100 m samples (Table S3), emerging as the fast endpoint of a continuous gradient. On the tailwind side alone the boys' 100 m contrast was small (0.071 versus 0.064 s, overlapping intervals) and the girls' larger (0.083 versus 0.057 s). The gradient's dominant mechanism is exposure time: within the registry, as within the collegiate sample, effects scaled approximately with race duration, across age and equally across ability within age, and the percentage effect was accordingly much flatter — though still declining. Across samples the proportionality is only approximate: per second of race time the collegiate and world-class 100 m slopes lie 15–27 % below the registry's, and at equal race times the registry's seniors gain a fifth to a quarter more than the slowest collegiate tertile, so no single per-second rate transfers between registries unchanged. The residual decline cannot be the aerodynamic share, which grows with speed and would push the percentage effect the other way; it is consistent with a fixed wind being a larger fraction of a slower runner's airspeed, and perhaps with children's higher surface-to-mass ratio. Headwinds cost sprinters more than tailwinds gave them, by about as much as drag physics predicts for the observed winds on the 100 m, by less on the 60 m, and by more on the 200 m, where the gauge covers only the straight; in a post-plan extension, class-specific short hurdles matched flat sprints over 60 m and mostly exceeded them over 100 and 110 m,

partly through longer race times.

4.2 Asymmetry, speed and the psychological interpretation

An asymmetry larger than an elite benchmark does not by itself point to a psychological effect. Lybbert and Zheng (2023) found that headwinds slowed US high-school sprinters more than a drag model for elite men predicts — head/tail ratios of 1.74 for boys and 2.08 for girls in their athlete fixed-effects estimates — and attributed the excess, larger for girls, to mindset. But the drag asymmetry itself grows as speed falls, roughly as $(2v+w)/(2v-w)$: over our observed 100 m winds, drag-only physics passed through the same estimator gives a ratio of 1.43 at a uniform elite 10.0 s, 1.57 at their boys' mean of 12.60 s and 1.69 at their girls' mean of 14.56 s, and more in samples of mixed speed, where the slowest runners carry the largest ratios (1.67 for our boys and men, averaging 12.08 s, and 1.73 for our girls and women, averaging 13.53 s). Where the benchmark is matched to the estimator and to each athlete's own speed, the 100 m asymmetry equals the drag prediction for girls and women and falls short of it for boys and men; the sex difference follows the speed difference, and the asymmetry is, if anything, smaller within meets (Table S8). Their boys' ratio is thus about what drag predicts for runners of their speed. Their girls' ratio remains above the 1.73 of our faster girls and women, but an unreported wind distribution and the absence of meet-level controls — headwinds often come with cold, wet conditions — leave physics and weather as candidate explanations for that remainder. Their further argument, that this psychological slack let a viral record shift subsequent times, rests on this premise. The external samples sharpen the point: collegiate sprinters, between the high-school and the elite samples in speed, match drag at their own speeds as the registry does, and only the championship finalists exceed it (1.9 against 1.45–1.49) — on the tailwind side, since a tailwind is worth least to them (0.036 and 0.044 s per m/s) while their headwind slopes (0.070 and 0.083) are about the collegiate ones (0.077 and 0.081). Selection on favorable readings at

the fast end, not a headwind penalty among the slow, is then the natural suspect before mindset.

4.3 Implications

For measurement, the results supply what has been missing: empirically grounded, age- and event-specific wind values for youth athletics, with a full correction table by event, sex and age band in Table S9. Since neither the absolute nor the percentage effect is strictly invariant, corrections should be age-graded; a single senior-derived factor understates youth wind effects by up to two-fifths. For fairness, the $\pm$2.0 m/s window does not favor the youngest uniformly. In seconds, a legal tailwind is worth about as much to boys aged 12–13 as to senior men and more to girls than to senior women; relative to athletes' own within-season variability it is worth less for the young than for seniors (on the 100 m, 0.67–0.68 of the robust scale at 12–13 against 0.90–1.26 at 20–34). What it does in every age band is move 100 and 200 m marks by more than a tenth of a second, and raw marks are what season lists, records and qualifying standards compare. The 2025 list exercise makes it concrete: still-air correction changes who is in the top 10, senior lists included. The practical conclusions are modest and do not require rule changes: report and store wind with youth results (as Norway does), prefer wind-adjusted or wind-stratified comparisons when marks inform selection or records, and treat single wind-assisted breakthroughs with proportionate skepticism. For talent evaluation the point compounds a broader lesson of the identification literature — early performance is an unreliable selection signal (Vaeyens et al., 2008; Guttormsen, 2026): at the ages where selection decisions lean hardest on raw times, a meter per second of wind moves those times by more seconds than at senior level.

4.4 Strengths and limitations

Strengths include scale (392 797 analyzed results, 295 293 of them identifying; 74 110 athlete-season-event cells), a within-athlete design that removes stable athlete and seasonal composition, direction-and-magnitude agreement with the elite literature at the senior endpoint and, with the same estimator, with collegiate and world-class samples (Table S3), internal consistency across age, ability, events and the hurdles extension, and a plan committed before estimation. Limitations are stated as read rules: gauge readings mismeasure athlete-experienced wind, so to the extent the error is classical the sprint estimates are lower bounds, and the meet-FE attenuation shows how much identification leans on between-meet wind variation; the gauge protocol differs by event family, and for the bend events measures only part of the course, so cross-family comparisons (and the low per-second rates of the 150/200 m) partly reflect measurement rather than physiology; wind is not randomized, and within-season form-weather covariation is controlled only by month, round and meet terms; jump estimates use the wind of the measured attempt only, so their bias direction is not established; the event-wide plausibility bounds admit a few gross recording errors, which barely move the slopes but motivated the robust scale in H5; the senior stratum is 20–34, so masters athletics is outside scope; missing wind (4.8 % of approved outdoor results, concentrated in the smallest meets) limits completeness but gives no mechanism for bias within athlete-seasons; temperature and other weather are unobserved (a meteorological linkage is a natural extension); and the setting is one federation's registry: the mechanisms are physical and the external samples reproduce the pattern, but not the per-second rate (Table S3).

5 Conclusion

Wind effects in athletics are not a constant to be borrowed from elite studies: the same estimator gives 0.048–0.059 s per m/s on the 100 m in collegiate and world-class samples against 0.082 in the registry. Per m/s of wind they are 34–70 % larger for child sprinters than for seniors, largely because slower athletes are exposed longer; they fall with ability for the same reason; and headwinds cost more than tailwinds give, by about as much as drag physics predicts on the 100 m. Within the legal window a +2.0 m/s tailwind is worth 0.11–0.25 s on the 100 and 200 m from age 12 — roughly the same across ages for boys and men, more for girls than for senior women — enough to reorder season lists at youth and senior ages alike. The remedy requires no new rules, because it operates on lists rather than races: wind is already recorded with every result, and the age-specific corrections estimated here (Table S9) can make lists, records and talent evaluation measurably fairer — for seniors too — while every race is run exactly as before.

Declarations

Acknowledgments. The authors thank the Norwegian Athletics Federation, whose systematic registration of results — wind readings included, down to the youngest classes — makes measurement research of this kind possible, and Tilastopaja Oy (Turku, Finland), which compiled the competition records of the championship finalists and permitted their use in this study.

Disclosure statement. The authors report there are no competing interests to declare. AGG maintains a non-commercial statistics platform built on the federation's public results database.

Funding. The authors received no funding for this work.

Declaration of generative AI use. The authors used Claude (Anthropic; Claude Opus 5.5, Fable 5 and 5.1, and Haiku 5.5, via Claude Code, September–October 2026) to write and check analysis code, to cross-check every reported number against the analysis output, and to draft and edit text. The authors reviewed and verified all code, output and text, confirm the originality and accuracy of the content, have checked the terms of use of the tool and confirm its suitability for publication, and take full responsibility for the integrity of the whole content, including the accuracy of the references.

Data availability statement. The competition records used in this study are publicly accessible via the Norwegian Athletics Federation's online register for ages 13 and above; records for the 10–12 age classes are not publicly released by the federation. The derived analysis dataset cannot be publicly shared because it links the records to athletes' birth years and therefore contains personal data under the EU General Data Protection Regulation (GDPR). The full analysis code, with the aggregate estimation files, will be deposited in a public repository upon acceptance. The external samples (Table S3) are drawn from the public athlete profiles of the Track & Field Results Reporting System (TFRRS; https://www.tfrrs.org) and from the subscription database Tilastopaja (Tilastopaja Oy, Turku, Finland), whose records are not redistributed.

References

Guttormsen, A. G. (2026). Longitudinal performance development in youth track and field: a population-level study. *Journal of Science and Medicine in Sport*. Advance online publication. doi:10.1016/j.jsams.2026.07.006.

Haugen, T. and Buchheit, M. (2016). Sprint running performance monitoring: methodological and practical considerations. *Sports Medicine*, 46(5):641–656.

Hollings, S. C., Hopkins, W. G., and Hume, P. A. (2012). Environmental and venue-related factors affecting the performance of elite male track athletes. *European Journal of Sport Science*, 12(3):201–206.

Lakens, D. (2022). Sample size justification. *Collabra: Psychology*, 8(1):33267.

Linthorne, N. P. (1994). The effect of wind on 100-m sprint times. *Journal of Applied Biomechanics*, 10(2):110–131.

Lybbert, T. J. and Zheng, G. (2023). Feeling fast? Beliefs and performance among high school sprinters. *Journal of Quantitative Analysis in Sports*, 19(3):153–170.

Moinat, M., Fabius, O., and Emanuel, K. S. (2018). Data-driven quantification of the effect of wind on athletics performance. *European Journal of Sport Science*, 18(9):1185–1190.

Moulton, B. R. (1986). Random group effects and the precision of regression estimates. *Journal of Econometrics*, 32(3):385–397.

Mureika, J. R. (2001). A realistic quasi-physical model of the 100 m dash. *Canadian Journal of Physics*, 79(4):697–713.

Vaeyens, R., Lenoir, M., Williams, A. M., and Philippaerts, R. M. (2008). Talent identifica-

456 tion and development programmes in sport: current models and future directions. *Sports*
457 *Medicine*, 38(9):703–714.

458 von Elm, E., Altman, D. G., Egger, M., Pocock, S. J., Gøtzsche, P. C., and Vandenbroucke,
459 J. P. (2007). The Strengthening the Reporting of Observational Studies in Epidemiol-
460 ogy (STROBE) statement: guidelines for reporting observational studies. *The Lancet*,
461 370(9596):1453–1457.

Tables and figures

Table 1: Analyzed events: volume, marks, and wind exposure (2011–2026).

Event	Results	In cells ≥2	Athlete-seasons	Athletes	Median mark	Wind, median (p5–p95)
60 m	103 286	72 558	19 354	10 790	9.59 s	+0.1 (−2.3–+2.5)
80 m	5 795	1 410	620	466	12.53 s	+0.0 (−2.6–+2.5)
100 m	76 430	64 557	13 222	5 451	12.88 s	+0.4 (−2.2–+2.8)
150 m	2 151	328	150	125	19.64 s	+0.3 (−2.5–+2.2)
200 m	76 113	59 246	15 549	6 608	27.99 s	+0.4 (−2.2–+2.9)
Long jump	51 866	35 774	8 126	4 365	4.42 m	+0.5 (−1.6–+2.7)
Long jump (zone)	54 665	44 336	12 338	7 835	3.64 m	+0.3 (−1.7–+2.7)
Triple jump	14 631	11 658	2 890	1 543	10.31 m	+0.7 (−1.3–+2.9)
Triple jump (zone)	7 860	5 426	1 861	1 413	8.42 m	+0.4 (−1.5–+2.8)

Results: analyzed rows (approved, outdoor, electronic, ages 10–34, wind within ±8 m/s). In cells ≥2: the estimation sample (the n of Table 2). Athlete-seasons and athletes: units contributing to estimation in that event. Counts and wind summaries are computed from the analysis file; the estimation files are available on request.

Table 2: Effect per m/s of wind (linear fit over head- and tailwinds, signed as a gain with tailwind), within athlete × season (95 % CIs clustered on meet).

Event	Per m/s, linear (95 % CI)	Percent per m/s (95 % CI)	n	Athlete-seasons
60 m	0.059 s (0.057–0.061)	0.626 (0.602–0.649)	72 558	19 354
80 m	0.075 s (0.062–0.088)	0.630 (0.531–0.729)	1 410	620
100 m	0.082 s (0.080–0.085)	0.646 (0.625–0.666)	64 557	13 222
150 m	0.081 s (0.045–0.118)	0.467 (0.278–0.656)	328	150
200 m	0.121 s (0.113–0.130)	0.442 (0.410–0.474)	59 246	15 549
Long jump	3.16 cm (2.86–3.46)	0.676 (0.604–0.747)	35 774	8 126
Long jump (zone)	2.11 cm (1.90–2.32)	0.570 (0.513–0.628)	44 336	12 338
Triple jump	4.18 cm (3.53–4.84)	0.394 (0.331–0.458)	11 658	2 890
Triple jump (zone)	3.88 cm (2.87–4.89)	0.449 (0.331–0.568)	5 426	1 861

Point estimates are identical under athlete clustering (the analysis plan's original choice), whose intervals are roughly half as wide in the large sprints and somewhat narrower elsewhere; both sets are in the estimation files available on request.

Table 3: Effect per m/s of wind (linear fit; seconds, 95 % CI clustered on meet) by age band and sex, sprints.

Event	Sex	10–11	12–13	14–15	16–17	18–19	20–34
60 m	Boys/men	0.070 (0.065–0.075)	0.063 (0.059–0.067)	0.051 (0.046–0.056)	0.041 (0.031–0.050)	—	—
60 m	Girls/women	0.064 (0.059–0.068)	0.056 (0.053–0.059)	0.053 (0.049–0.057)	0.049 (0.037–0.061)	—	—
100 m	Boys/men	—	0.101 (0.084–0.117)	0.090 (0.085–0.095)	0.079 (0.074–0.083)	0.078 (0.072–0.083)	0.071 (0.066–0.075)
100 m	Girls/women	—	0.097 (0.086–0.107)	0.095 (0.089–0.100)	0.082 (0.077–0.087)	0.081 (0.075–0.087)	0.072 (0.067–0.078)
200 m	Boys/men	0.144 (0.083–0.204)	0.142 (0.121–0.163)	0.125 (0.112–0.139)	0.112 (0.100–0.125)	0.101 (0.088–0.115)	0.085 (0.068–0.101)
200 m	Girls/women	0.094 (0.050–0.137)	0.140 (0.121–0.159)	0.133 (0.120–0.145)	0.123 (0.106–0.141)	0.096 (0.082–0.109)	0.095 (0.075–0.115)

Dashes: fewer than 300 results in 100 athlete-seasons, the reporting threshold (60 m above 17, which is rarely run outdoors; 100 m at 10–11, where 60 m is the standard sprint). Table values are reproduced from the estimation files available on request.

Table 4: Advantage of a +2.0 m/s tailwind over still air, valued from the slope over 0 to +2 m/s: seconds, and in parentheses the share of the athlete's robust within-season scale of wind-adjusted performance.

Event	Sex	10–11	12–13	14–15	16–17	18–19	20–34
60 m	Boys/men	0.11 s (0.64)	0.13 s (0.90)	0.11 s (0.96)	0.09 s (0.99)	—	—
60 m	Girls/women	0.11 s (0.66)	0.11 s (0.83)	0.09 s (0.76)	0.06 s (0.59)	—	—
100 m	Boys/men	—	0.16 s (0.67)	0.15 s (0.91)	0.14 s (1.03)	0.16 s (1.25)	0.15 s (1.26)
100 m	Girls/women	—	0.15 s (0.68)	0.15 s (0.82)	0.13 s (0.86)	0.14 s (1.04)	0.11 s (0.90)
200 m	Boys/men	0.27 s (0.37)	0.16 s (0.29)	0.16 s (0.41)	0.19 s (0.66)	0.13 s (0.47)	0.14 s (0.53)
200 m	Girls/women	0.51 s (0.62)	0.25 s (0.45)	0.20 s (0.47)	0.18 s (0.51)	0.16 s (0.50)	0.14 s (0.52)

Seconds are 2× the slope over 0 to +2 m/s from a piecewise model with separate slopes for headwinds and for tailwinds above +2 (Table S9 gives the slopes). The robust scale is 1.4826 × the median absolute deviation of within-cell residuals of wind-adjusted marks over cells with at least three results, per event × sex × band — a denominator largely free of the wind-driven variance itself, of singleton-cell bias and of gross recording errors. Values reproduced from the estimation files available on request. The 60 m rows at 16–17 and the 200 m rows at 10–11 rest on thin samples.

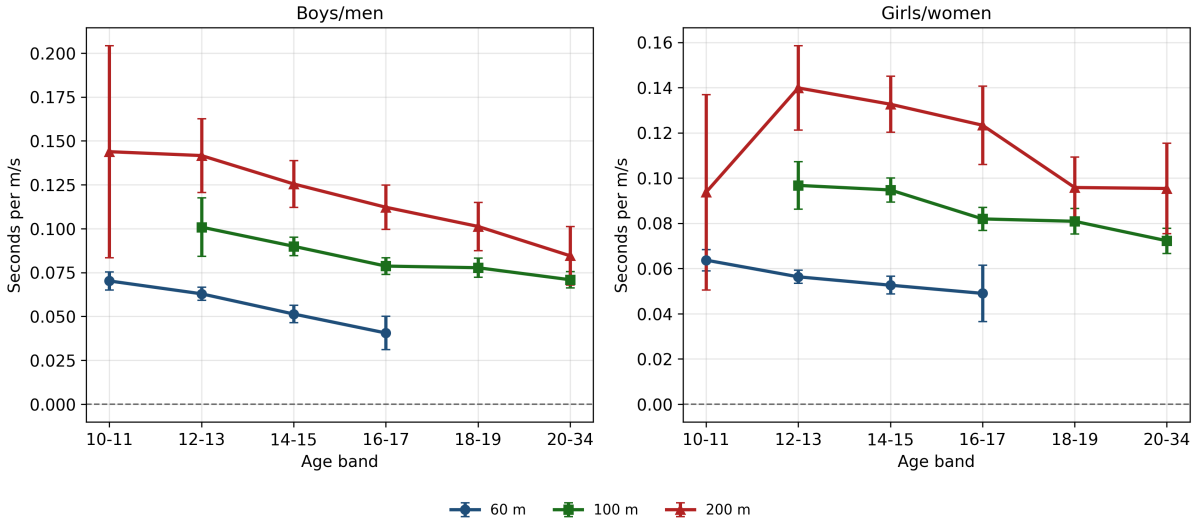

Boys/men
Girls/women
Seconds per m/s
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
10-11
12-13
14-15
16-17
18-19
20-34
Age band
60 m
100 m
200 m

Figure 1

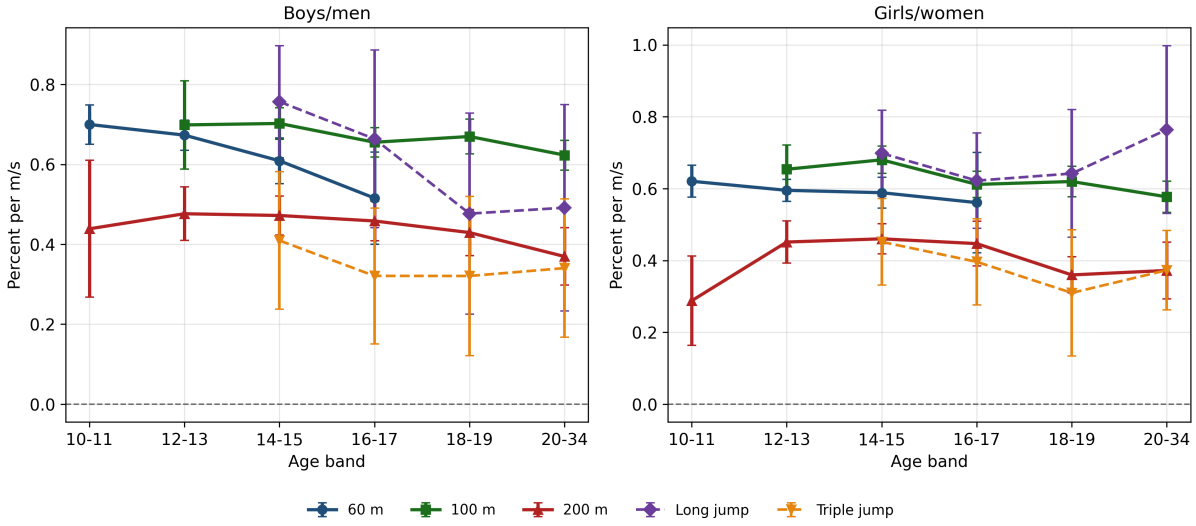

Boys/men
Girls/women
Percent per m/s
Age band
10-11
12-13
14-15
16-17
18-19
20-34
0.0
0.2
0.4
0.6
0.8
1.0
60 m
100 m
200 m
Long jump
Triple jump

Figure 2

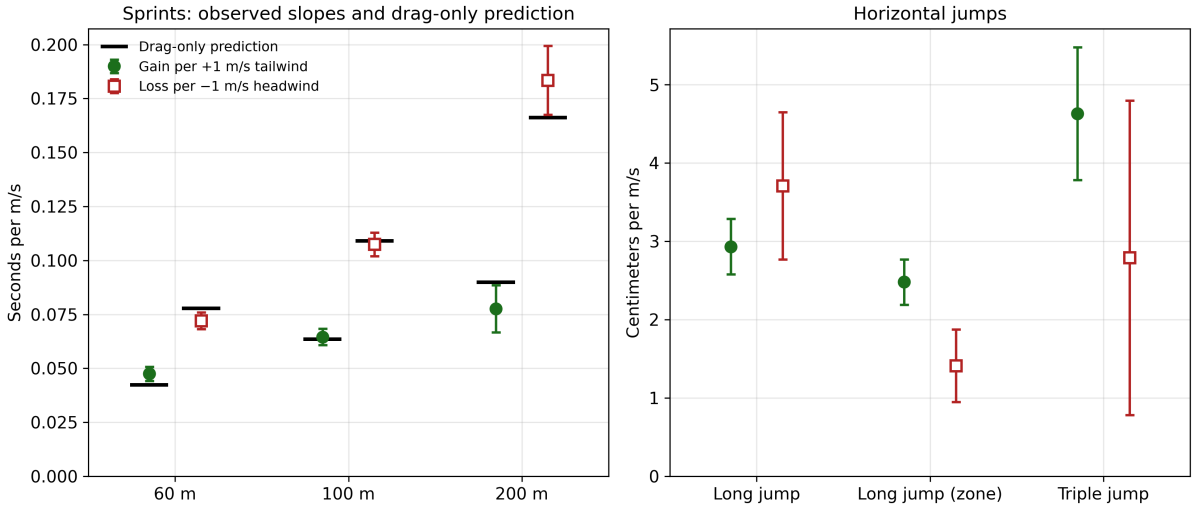

Sprints: observed slopes and drag-only prediction
Drag-only prediction
Gain per +1 m/s tailwind
Loss per −1 m/s headwind
Seconds per m/s
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
60 m
100 m
200 m
Horizontal jumps
Centimeters per m/s
5
4
3
2
1
0
Long jump
Long jump (zone)
Triple jump

Figure 3

Figure captions

Figure 1. Effect per m/s of wind (linear fit over head- and tailwinds) by age band, in seconds, for sprints (within athlete × season; 95 % CIs clustered on meet).

Figure 2. The same scaling in percent of performance, for sprints (solid lines) and horizontal jumps (dashed). For the sprints the percentage effect is far flatter across age than the absolute effect, but still declines; the jumps show no consistent age direction (95 % CIs clustered on meet).

Figure 3. Separate head- and tailwind slopes (within athlete × season; points with 95 % CIs clustered on meet; filled circles tailwind, open squares headwind). Black bars: the slopes that drag-only physics predicts for the observed winds (sprints), with the drag response scaled to the observed linear slope.